

DONGCHEN XIE

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🎓 EDUCATION BACKGROUND

Renmin University of China School of Information Information Security 2023 – Present
Advisor: Wei You Lab: Intelligent Software Security Research Group

University of Electronic Science and Technology of China
School of Information and Software Engineering Internet Security 2019 – 2023
GPA: 3.96/4.0 Weighted Average: 90.29 Rank: 10/160 (Top 6%)

★ RESEARCH ACHIEVEMENTS

BRIDGEROUTER: Automated Capability Upgrading of Out-Of-Bounds Write Vulnerabilities to Arbitrary Memory Write Primitives in the Linux Kernel *IEEE S&P, 2025*

Dongchen Xie, Dongnan He, Wei You, Jianjun Huang, Bin Liang, Shuitao Gan, Wenchang Shi.

- This paper proposes a practical **kernel OOB vulnerability exploitation technique** that extends the limited capability of kernel OOB vulnerabilities to support arbitrary memory writes, further enabling privilege escalation and other attacks.
- Designed and implemented a prototype system, improving static analysis tools to filter key objects, integrating static instrumentation and fuzzing to trigger object allocation, and automating the matching of vulnerabilities and key objects for exploitation.
- Conducted a comprehensive evaluation on the accuracy of bridge and router object identification in the Linux kernel, the applicability across memory allocation caches, and the effectiveness of real-world vulnerability exploitation.

Define-Use Guided Path Exploration for Better Forced Execution *ACM SIGSOFT ISSTA, 2024*

Dongnan He, **Dongchen Xie**, Yujie Wang, Wei You, Bin Liang, Jianjun Huang, Wenchang Shi, Zhuo Zhang, Xiangyu Zhang.

- This paper proposes a novel and practical **path exploration strategy**, focusing on def-use relationship coverage in target binaries.
- Responsible for experiment design and testing: Set up experimental environments using QEMU and Pin-Tools under Linux and Windows, monitored system call coverage and parameter variations, and analyzed anomalous behavior during fuzzing.
- Evaluated the approach using the SPEC2000 benchmark suite (12 programs of different scales) and 200 recent real-world malware samples (100 on Linux, 100 on Windows). Experimental results demonstrate superior performance in memory dependency coverage and malware behavior exposure compared to existing forced execution tools.

🏆 HONORS AWARDS

National College Mathematics Competition	<i>First Prize</i>	January 2021
Computer Works Competition	<i>Provincial Second Prize</i>	June 2021
Outstanding Student Scholarship	<i>First Prize</i>	Every Academic Year(2019-2024)
Enterprise Scholarship	<i>First Prize</i>	Every Academic Year(2019-2022)